HOLY ANGEL UNIVERSITY
SCHOOL OF COMPUTING

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Answer the following questions:

- How do you differentiate a system memory and a mass storage system.
- Describe briefly and differentiate types of a system memory.
- How do you differentiate HDD and SSD, what factors should we consider when selecting mass storage?
- Identify the different options available for external storage.

How do you differentiate a system memory and a mass storage system.

System memory and mass storage are essential components of a computer system, each serving a specific purpose. RAM, or system memory, offers temporary storage for data required by the CPU during active activities, but it is volatile, which means it loses data when the computer is shut down or restarted. Mass storage, on the other hand, is used for long-term data storage, such as the storage of the operating system, software, documents, and media assets. Mass storage is slower than RAM because it uses mechanical or electrical procedures to access data. Mass storage devices have greater storage capabilities, often measured in terabytes or more, making them appropriate for long-term data storage. Both components are important for a computer's overall operation.

Describe briefly and differentiate types of a system memory.

System memory, also known as RAM, exists in various types, and (double data rate, fourth generation random access memory) DDR4 is the prevailing and contemporary selection for typical computing tasks. (Double Data Rate Version 3) DDR3 remains relevant in older computer systems, while (Double Data Rate Version 2) DDR2 and (Double Data Rate) DDR are outdated options. LPDDR (Low Power Double Data Rate) is tailored for mobile devices. (Error correction code memory) ECC memory offers error correction for data reliability, and registered memory is employed in server environments. SDRAM (Synchronous Dynamic Random Access Memory) is a type of computer memory that allows for quick and synchronized data writing, resulting in better overall system performance. Static Random-Access Memory (SRAM) is a volatile semiconductor used in computers and electronic devices for high-speed data storage and retrieval, providing quicker, more power-efficient storage in flip-flop circuits than dynamic RAM. Lastly, DRAM is a volatile computer memory that stores data on integrated circuits and requires constant refreshing for data integrity. It is commonly utilized because of its great storage density and low cost. The decision on which type of system memory to use depends on the specific requirements, compatibility, and functionality of the computer system.

How do you differentiate HDD and SSD? What factors should we consider when selecting mass storage?

(hard disk drive) HDDs and (solid-state drive) SSDs are two types of hard disk drives that are used to store data. HDDs use spinning magnetic disks to access data, whereas SSDs employ NAND flash memory for storage. SSDs provide advantages such as increased speed, durability, decreased power usage, less noise and heat, and a smaller form factor. HDDs, on the other hand, are bigger and heavier. Considering aspects like performance requirements, storage capacity, budget, usage situations, gaming and multimedia preferences, mobility, data dependability, and security when picking between HDD and SSD, due to the lack of moving components, SSDs are more dependable, making them a favored choice for crucial data storage. Hybrid drives (SSHDs) provide a middle ground by combining SSD performance with HDD capacity.

Identify the different options available for external storage.

External storage options provide additional storage capacity for computers and devices. These options encompass a variety of choices, including external HDDs, external SSDs, USB flash drives, memory cards and SD cards, NAS devices, external optical drives, cloud storage services, external RAID enclosures, docking stations, and wireless external drives. HDDs are portable and offer substantial storage space, while SSDs offer high-speed data transfer. USB flash drives are compact and lightweight, and memory cards and SD cards serve as removable storage for various data types. NAS devices are dedicated networked storage systems, and external optical drives are used for reading and writing optical discs. Cloud storage services offer remote storage solutions, while RAID enclosures house multiple drives for performance and redundancy. Docking stations facilitate the connection of internal drives externally.